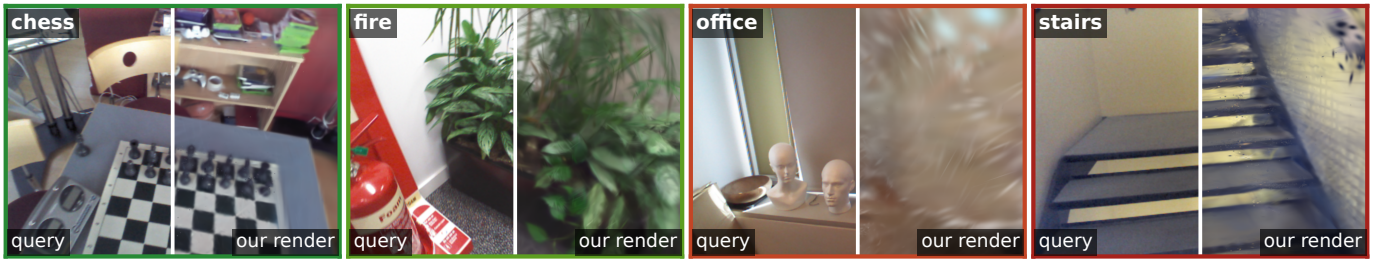
USGLoc returns a pose and its SE(3) covariance — not just a point.

The reported σ ranks the method's own answers: small σ → the render matches the query photo; large σ → it visibly does not.



σ = **0.22 cm** error 0.14 cm σ = **1.60 cm** error 0.40 cm σ = **119 cm** error 16.7 cm σ = **1160 cm** error 144 cm

Panel: left half = query photo, right half = 3DGS render at our pose.
Bar = reported σ on one shared log scale (0.1 cm – 20 m).
Black notch = actual error, on that same scale.
Panels ordered left → right by reported σ.

Does a method's own confidence rank its errors?

USGLoc σ	**43×**	(0.28 → 12.1 cm)
ACE inliers	**7×**	(0.47 → 3.3 cm)

median error, top vs bottom decile; 500 queries, 5 scenes